



Mohamed Tawhid

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Summary

Results-driven **Electrical power and Energy Engineer** specializing in the design, operation, and optimization of renewable energy systems (up to 1000 kW). Proven expertise in managing **10+ international deployments** of energy systems for **UNDP, Siemens, and Schneider Electric**. Secured funding for **8 projects** by developing compelling business cases. Passionate about leveraging data analysis tools (**Python, R, Power BI**) and Machine Learning to drive the **zero-carbon energy transition**.

Education

Erasmus Mundus Joint Master's Full Scholarship – Energy Systems Optimization.

- **Semester 4:** Energy Data Space Internship at ENTSO-E.
- **Semester 3:** Autonomous University of Barcelona. [GPA: 4.0/4.0] (Energy Systems Optimization)
- **Semester 2:** Vienna University of Technology. [GPA: 4.0/4.0] (Numerical – Modelling Training).
- **Semester 1:** Università degli Studi dell'Aquila. [GPA: 3.0/4.0] (Mathematical Modeling).

Sep 2024 - Aug 2026

Brussels, Belgium
Barcelona, Spain
Vienna, Austria
L'Aquila, Italy

Bachelor of Science – Electrical Power and Renewable Energy Engineering. Ain Shams University.

- **Full Scholarship with GPA: 3.42/4.0** (Dean's List of Honors). Rank: **Top 5%**.
- **Thesis Title:** "Towards Efficient Energy Usage at Ain Shams University Campus":
 - Designed **2 MW** PV systems by **PVsyst**, reducing campus CO₂ emissions by **750 tons/year**.
 - Optimized energy yield through **PVSOL**-powered 3D shading analysis and layout optimization.
 - Increased renewable energy penetration by **15%** across **11 buildings**, generating **3.1 GWh** annually.

Aug 2019 – Jul 2023

Cairo, Egypt

Semester Abroad - Energy Engineering. University of Nebraska-Lincoln.

- **Full Scholarship with GPA: 3.75/4.0** (Dean's List of Honors).
- **Capstone Project:** Wind Farm Design to Provide Zero-Carbon Energy to Hettinger City:
 - Engineered 6 MW wind farm to cover the city's total energy demand by producing 32.8 GWh annually.
 - Performed financial modeling to calculate the LCOE, demonstrating the ability to supply power 9% below the national average at \$0.1026/kWh, proving the project's economic viability.

Aug 2022 – Dec 2022

Nebraska, USA

Work Experience

Energy Data Space Engineer | ENTSO-E.

- Applied machine learning for **time-series forecasting** on Germany's 2024 **hourly load data** to generate actionable insights for grid stability and capacity planning.
- Transformed raw **data** from **172 energy meters** into strategic **business intelligence**, demonstrating capacity for optimizing grid operations and asset performance.
- Identified technical gaps and opportunities for TSOs in the Common European Energy Data Space.
- Evaluated data space governance models and technical designs to propose best practices.

Feb 2026 – Aug 2026

Brussels, Belgium

Green Energy Engineer | United Nations Development Program.

- Managed technical operations and daily monitoring for a portfolio of **10 international** solar PV Installations (**20-1000 kW**), ensuring **98%** system availability and maximizing energy output.
- Modeled and optimized hybrid energy systems using **HOMER Pro** and PVSol, conducting techno-economic analyses that reduced **LCOE** by an estimated **15%** and increased **Renewable Fraction** by **40%** in proposal.
- Authored **8** comprehensive business cases for solar PV, e-mobility, and energy efficiency projects, successfully securing **\$ 200,000+** funding for UN offices.

Feb 2024 – Jul 2024

Copenhagen, Denmark

Electrical Design Engineering Program | Shaker Consultancy Group

- Designed **15 MW** of electrical distribution infrastructure, including three **11kV/0.4kV** transformer and kiosk substations, ensuring reliable power supply for commercial and residential clients.
- Modeled energy distribution networks and conducted protection coordination studies using **ETAP**, ensuring **100%** compliance with IEC 61439 standards and guaranteeing system safety and reliability.
- Executed short-circuit analysis for new electrical designs, mitigating risk and enhancing system reliability.

Sep 2023 – Dec 2023

Cairo, Egypt

Power Systems Engineer (Intern) SIEMENS	Jun 2022 – Sep 2022
- Performed optimal power flow analysis on a 6-bus test grid using ETAP and PSS®E, identifying configurations that reduced power losses by 8%. - Co-designed a feasibility study for a microgrid integrating 6.4 MW solar PV system and 12 MW wind farm. - Designed a 60-kW grid-tied commercial PV system using PVsyst; performed financial modeling that demonstrated a reduction in payback period from 8 to 5 years, increasing feasibility and client ROI.	Cairo, Egypt
Energy Management Engineer (Intern) Schneider Electric	Jan 2022 – Jun 2022
- Led the design and deployment of an Intelligent Motor Control Center educational kit for 500 students, providing real-time monitoring and protection of 3-ph motors by configuring TeSys T controller. - Simulated a 4000 A LV Switchboard, ensuring 100% compliance with IEC 61439 standards.	Cairo, Egypt

Leadership & Volunteering Experience

UN Graduate Study Programme Participant and Egypt Representative United Nations Office at Geneva.	Jun - Jul 2026
- Drafted policy recommendations on AI governance and emerging technologies through interdisciplinary research with UN agencies (ILO, OHCHR, ITU, WHO, UNHCR) and thematic Working Group collaboration. - Collaborated with an international cohort of 56 graduate students from 42 countries, engaging with UN officials, diplomats, and tech experts on digital cooperation, humanitarian innovation, and international security. - Explored multilateral governance via guided tours of the Palais des Nations and CERN, attendance at the AI for Good Global Summit, and diplomatic networking events with EU and Geneva representatives.	Geneva, Switzerland
EULiST Student Conference Representative IMT Nord Europe.	Jul 2025
- Led my team in securing 1st Place in the EULiST Green Hackathon by developing a proposal for a low-cost, student-led recycling system to reduce university carbon footprints. - Delivered a keynote presentation on global solar energy deployment, translating technical project experience from my UNDP role into strategic insights on community empowerment and energy transition.	Douai, France
University Delegate and Representative EULiST European University Alliance.	Dec 2024
Selected to represent at the 2nd international EULiST Conference, engaging with 200+ students and faculty from 10 European institutions to foster climate-neutral cities and universities.	Hannover, Germany
Student Ambassador USAID Scholars Activity.	Apr 2021 – Sep 2023
- Negotiated a partnership with management at the Benban Solar Park, securing 4 competitive engineering internships at Voltalia for fellow scholars, providing crucial industry exposure. - Represented the USAID scholars at highest administrative meetings with USAID Seniors.	USA – Egypt
Madagascar Solar Project Volunteer Engineers Without Borders.	Sep 2022 – Dec 2022
Co-developed a comprehensive technical manual for the installation and maintenance of photovoltaic systems to provide reliable electricity to 6 remote schools in Madagascar.	Nebraska, USA

Core Competencies & Technical Skills

- Renewable Energy Systems (PV, Wind) | Microgrid Design & Optimization | Business Case Development | Grid Integration | Data Analysis & Visualization | Project Management | Feasibility Studies | Electrical Design (IEC Standards).
- Electrical Design:** ETAP, PSS/E, AutoCAD. **Energy Simulation:** PVsyst, PV*SOL, HOMER Pro, RETScreen.
- Programming & Data:** Python (Pandas, NumPy, Machine Learning), MATLAB, R, Power BI, Advanced Excel, Microsoft 365.

Languages

- Arabic** (Native)
- English** (Advanced)(C1)
- German** (Pre-intermediate)(A2)
- French** (Beginner)(A1)

Achievements & Honors

- Secured **1st Place** in the **EULiST Green Hackathon** for my proposal of a low-cost, student-led campus recycling system. (2025)
- Achieved **3rd Place** internationally among 30 teams in the **Net Zero Carbon Campus** competition by Kingston University. (2023)
- Won **1st Place** in the Collaborative Design Challenge by developing an innovative engineering solution. (2022)
- Awarded a competitive **USAID** scholarship for a semester abroad at the University of Nebraska Lincoln. (2022)
- Won **1st Place** among 12 teams in a Schneider Electric design initiative for my electrical panel design. (2021)
- Awarded a full undergraduate scholarship from **USAID & MIT** for academic excellence and leadership potential. (2019-2023)